

August 10, 2026

Editor
The R Journal

Dear Editor,

Please consider our article titled “Stabilitest: Robustness and Fragility Analysis of Statistical Test Conclusions in R” for publication in the R Journal.

The manuscript introduces **stabilitest**, a package that asks how easily a statistical conclusion could be overturned. It combines three complementary sensitivity views of one pre-specified analysis — jackknife leave-one-out influence, greedy worst-case observation removal in the spirit of the maximum influence perturbation of Broderick, Giordano and Meager, and bootstrap same-decision rates — across two-sample location and proportion tests, linear model and ANCOVA terms, GLM and Cox terms, and TOST equivalence and non-inferiority endpoints.

We believe readers of the R Journal will find the article useful for two reasons beyond the software itself. First, the observed-sample perturbation analysis complements the broader emphasis on sensitivity analysis in ICH E9(R1). The article also positions the package carefully against the existing R ecosystem — **fragility**, **sensemakr**, **konfound**, **EValue**, **influence.ME**, **car**, and **zaminfluence** — distinguishing tools that reason about unmeasured threats from those, like this one, that interrogate the observed sample.

Second, the article takes an unusual position on interpretation thresholds. A categorical robustness verdict is treated as a claim requiring evidence: the package emits one only where an independent, pre-registered calibration study has validated thresholds for that exact analysis configuration, and otherwise suppresses the label while retaining all numeric output. We report a successful Fisher exact-test study, a prospective Welch study whose frozen training phase found no candidate satisfying the pre-specified gates after 4,500 completed analyses, and three negative ANCOVA studies. Because the Welch study failed closed at training, its held-out validation data were not opened. These results establish empirical infeasibility under their frozen designs without claiming that all possible future designs must fail. We consider the negative results contributions rather than omissions and welcome reviewer scrutiny of that framing.

All results in the article are computed from code or read from committed artifacts with committed generation scripts; nothing is transcribed. Heavy calibration is exposed through those scripts and frozen artifacts, while the article itself knits to HTML and PDF well inside the journal’s ten-minute budget. Full regeneration and a clean-checkout rehearsal are documented in **REPRODUCE.md** and the accompanying reproduction report.

The article targets the published CRAN package version 0.6.0 at <https://CRAN.R-project.org/package=stabilitest>. Final package and article checks are recorded in the accompanying reproduction report.

Regards,

Marius Ardelean
Independent researcher
marally@gmail.com